Model Development Phase Template

Date	25 February 2026
Team ID	259422
Project Title	Online Payments Fraud Detection using Machine Learning
Maximum Marks	5 Marks

Feature Selection Report Template

The feature selection process was based on:

Domain relevance to fraud behavior

Correlation analysis

Multicollinearity assessment

Feature importance from tree-based models

Risk of data leakage.

Feature	Description	Selected (Yes/No)	Reasoning
step	Represents time step in hours	Yes	Captures temporal transaction patterns; may indicate fraud bursts.
type	Transaction category (TRANSFER, CASH_OUT, etc.)	Yes	Strong association with fraud; TRANSFER highly correlated with fraudulent activity.
amount	Transaction amount	Yes	Major fraud indicator; fraudulent transactions often involve higher amounts.
oldbalanceOrg	Sender balance before transaction	Yes	Helps identify suspicious full-balance transfers.
newbalanceOrig	Sender balance after transaction	Yes	Detects sudden balance depletion patterns linked to fraud.
oldbalanceDest	Receiver balance before transaction	Yes	Useful for anomaly detection in fund transfers.
newbalanceDest	Receiver balance after	Yes	Helps measure abnormal balance

	transaction		jumps.
isFraud	Fraud label (target variable)	Yes (Target Only)	Dependent variable for supervised learning; excluded from feature matrix.
nameOrig	Sender account ID (if present)	No	Unique identifier; no predictive value; risk of overfitting/data leakage.
nameDest	Receiver account ID (if present)	No	Unique identifier; high cardinality; no generalizable predictive power.